

WORK IN PROGRESS

# Free-body diagrams for neural networks

Which internal statistics support  
further learning?



Neural  
networks  
have known  
microscopic  
laws



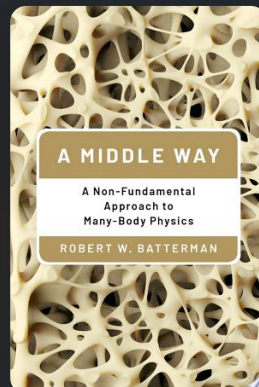
we do not  
understand  
how they  
compute—or  
what their  
representations  
mean



study  
dynamics  
to see  
how they  
learn

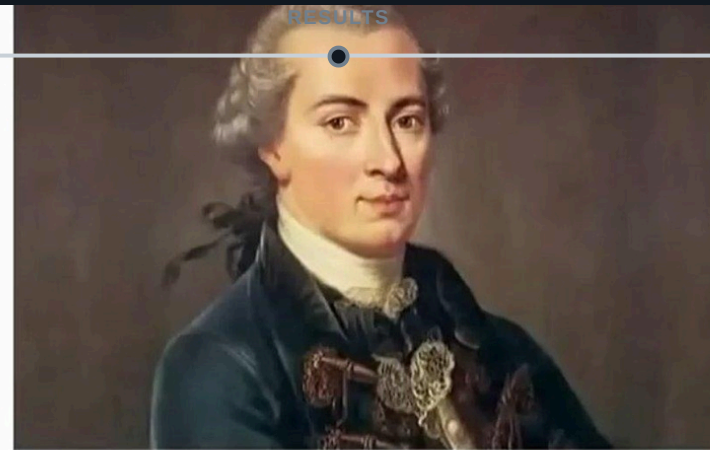
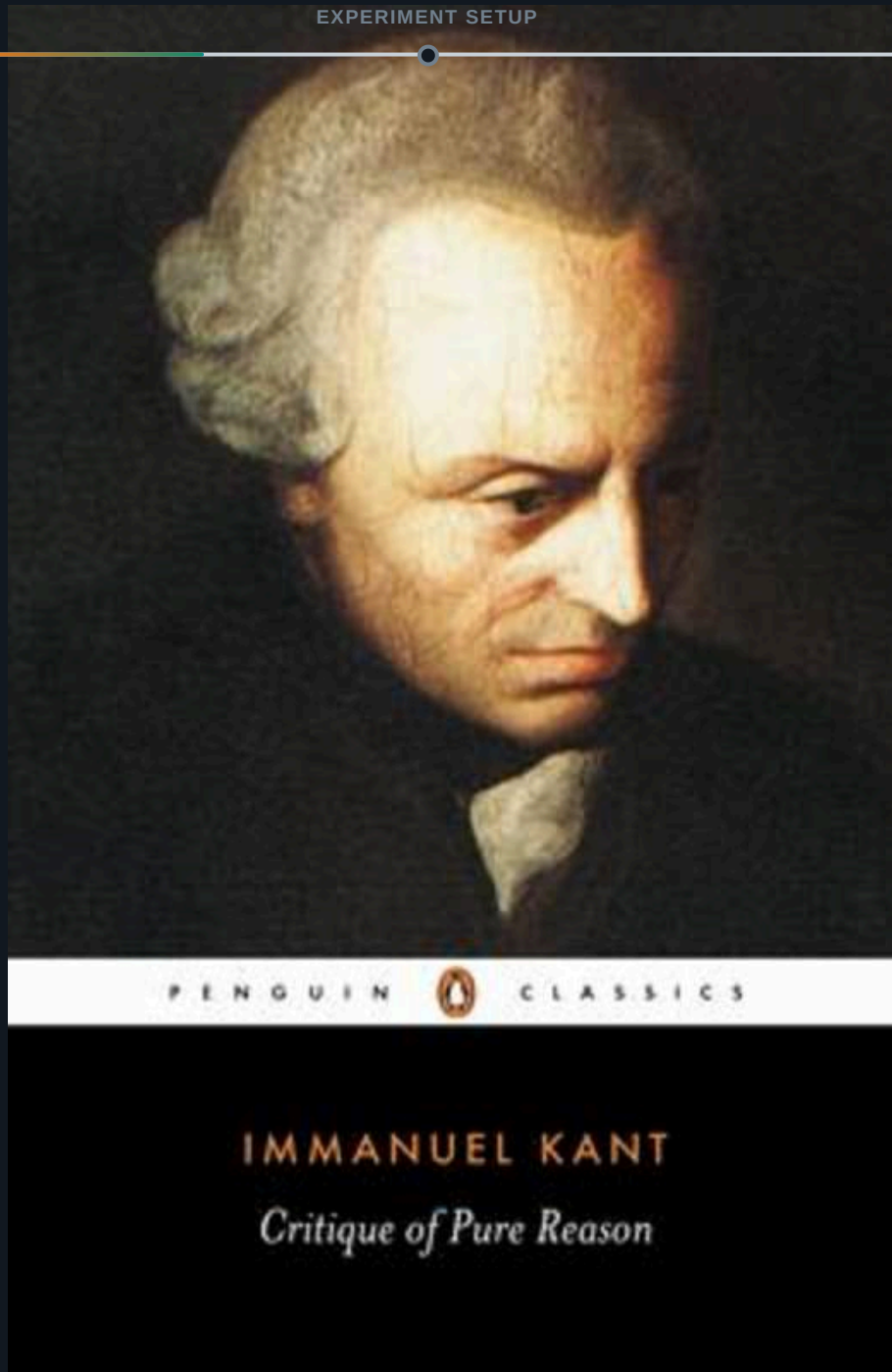


let  
dynamics  
tell us the  
natural  
ontologies



▶ Asymptotics, Minimal Models, Multiscale Techniques

▶ Methodology, Models, and Mathematics



## Immanuel Kant

Kant never left his home town, Koenigsberg (today's Kaliningrad), never married, never changed his daily schedule or his diet, and died, presumably happy and mildly bored, at the age of 80. His last words were: "It's fine."

## Bros Life:

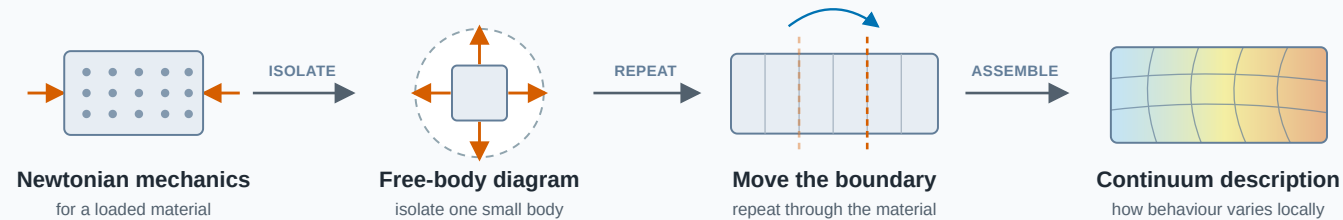


# Free body diagrams for neural networks

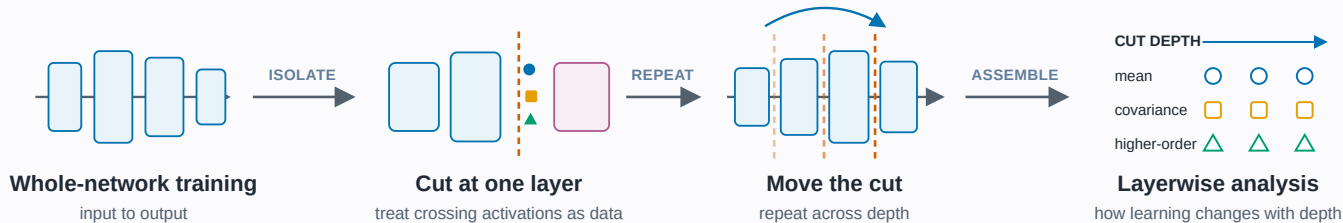
## From a free-body diagram to a layerwise picture

Use a local cut to understand a large system, then repeat the measurement across the system.

### MATERIALS AND MECHANICS



### NEURAL NETWORKS



● This is a methodological analogy: it organises measurements; it is not a claim that neural networks obey continuum mechanics.

# two networks: CNN and ResNet-18

trained on CIFAR

## The two networks and where we cut them

SCHEMATIC · NOT TO SCALE

Orange circles mark the activations saved for the later-part training experiment.

### Four-block residual CNN

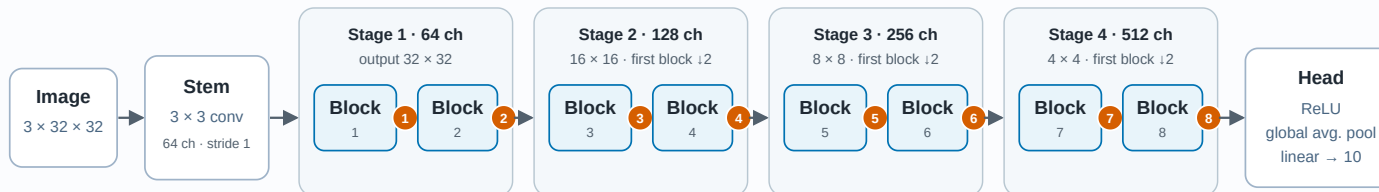
CIFAR-10 · cuts after blocks 1–4



Each block has two 3 × 3 convolutions with GroupNorm, a residual skip, and ReLUs.

### Normalization-free pre-activation CIFAR ResNet-18

cuts after all eight residual blocks



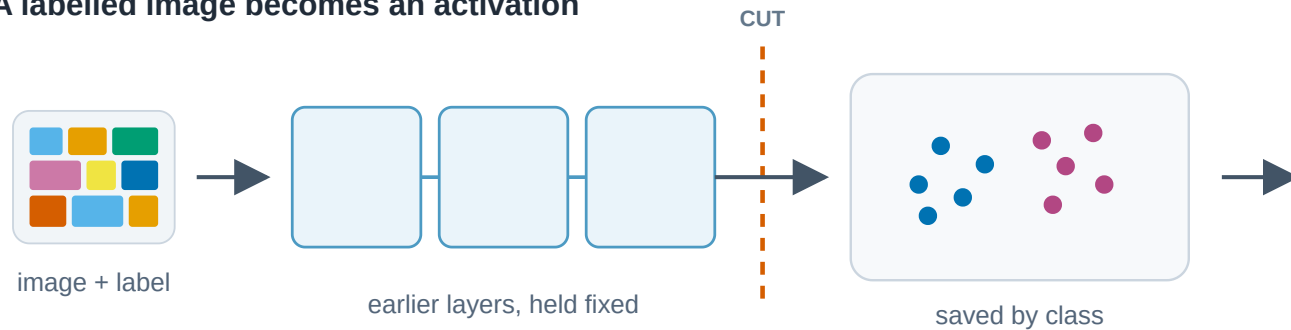
Each residual block is ReLU → 3 × 3 conv → ReLU → 3 × 3 conv, added to a skip path; there is no normalization.

At stage changes, the first block and its skip path downsample; the skip uses a 1 × 1 projection.

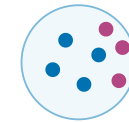
## Make datasets from one internal layer

Hold the earlier layers fixed, save what crosses the cut, then simplify those activations in two ways.

### A labelled image becomes an activation



### Three versions of the same internal dataset



#### Real activations

Use the saved examples themselves.



#### Fitted Gaussian

Match each class mean and covariance in the fitted PCA space.

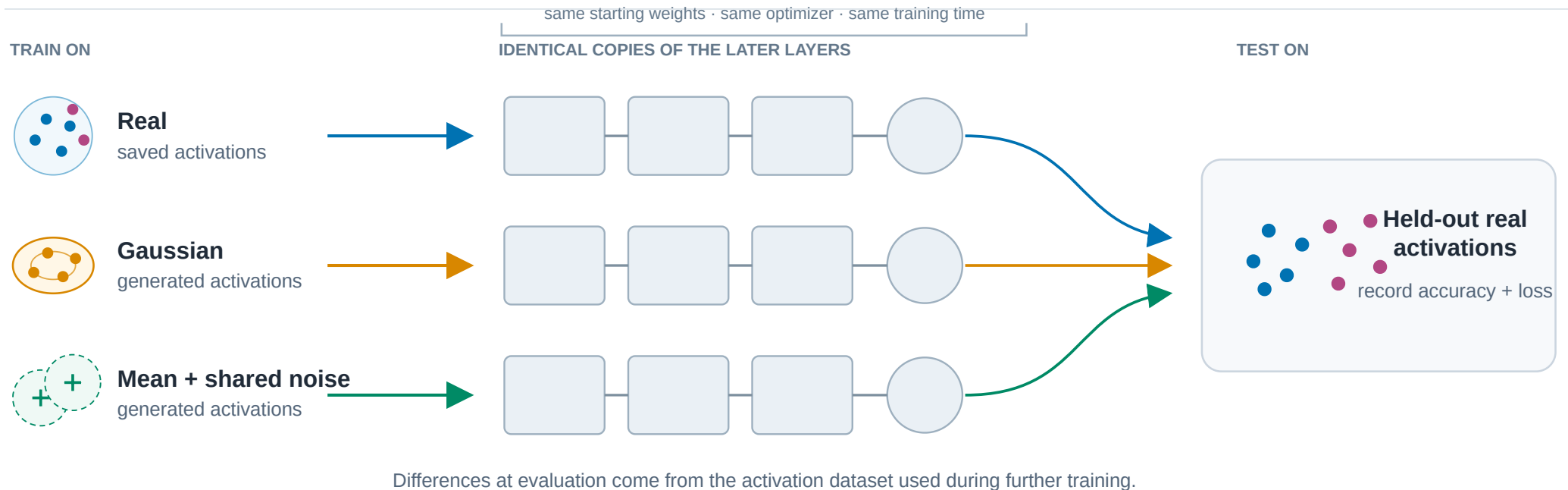


#### Mean + shared noise

Keep class means, but use the same noise scale for every class.

## Change only the data used for further training

The three copies start alike and get the same training budget. All are evaluated on real held-out activations.

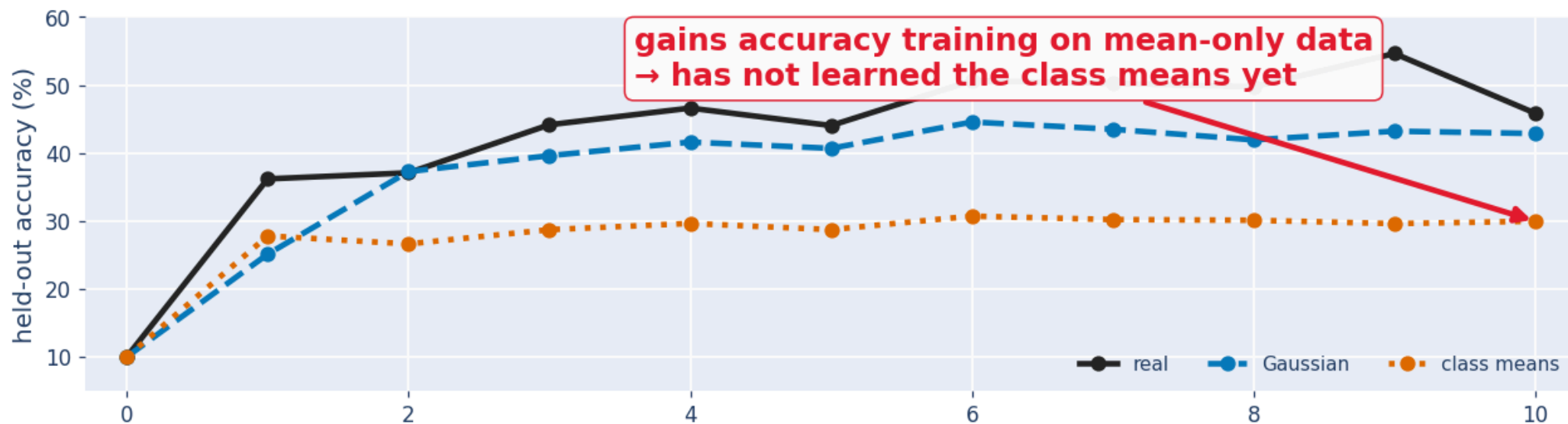




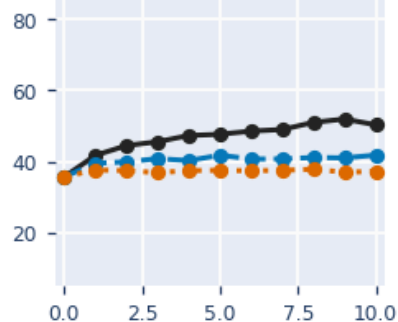
GIF

## RELAX SUFFIX WITH FROZEN PREFIX

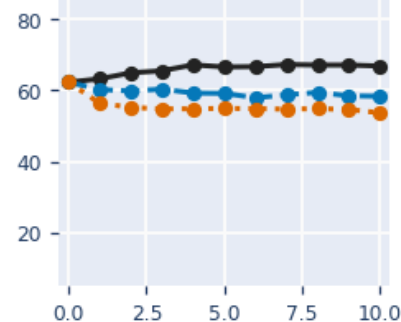
## ORDINARY-TRAINING EPOCH 0



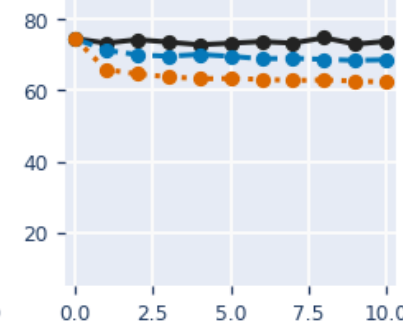
ordinary-training epoch 1



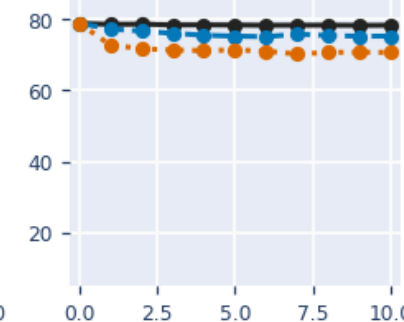
ordinary-training epoch 5



ordinary-training epoch 10



ordinary-training epoch 20



ordinary-training epoch 30

